Learning Platform Performance Analysis Using MySQL

Objectives

An online learning platform sells various digital courses to learners across different countries. Each

learner can purchase multiple courses, and each course belongs to a specific category.

The management team wants to analyze purchase data to understand:

- Sales trends
- Learner behavior
- Popular course categories

Your task is to use MySQL to design, query, and extend a relational database to derive meaningful

insights from this e-learning purchase data.

Dataset Description:

You will work with the following three tables:

Table: *learners*

Column	*Description*
learner_id	*Primary Key*
full_name	*Learner name*
country	*Country of residence*

Table: *courses*

Column	*Description*
course_id	*Primary Key*
course_name	*Course title*
category	*Course category*
unit_price	*Price per course*

[redacted]

Table: *purchases*

Columns	***Description***
purchase_id	*Primary Key*
learner_id	*Foreign Key → learners*
course_id	*Foreign Key → courses*
quantity	*Number of courses purchased*
purchase_date	*Date of purchase*

1. Database Setup & Data Entry

- Create a new database.
- Create the three tables with proper:
  - Primary keys, Foreign keys, Appropriate data types
- Insert sample data:
  - 4–5 learners
  - 4–5 courses (different categories)
  - 6–8 purchase records

learners_id	learners_name	country_name
1	anith	INDIA
2	arun	USA
3	kavya	INDIA
4	john	LONDAN
5	anu	INDIA
NULL	NULL	NULL

learnears 14

courses_id	courses_name	category	unit_price
1	digital marketing	marketing	5999.00
2	full stack course	programming	9999.00
3	power bi quary	data analytics	6999.00
4	python	programming	9999.00
5	excel advanced	productivity	3999.00
6	sql advanced	data analytics	5999.00
NULL	NULL	NULL	NULL

purchases_id	learnears_id	courses_id	quantity	purchases_date
1001	1	1	1	2025-02-12
1002	1	2	1	2025-02-11
1003	2	3	1	2025-02-17
1004	2	5	1	2025-02-19
1005	3	4	1	2025-02-25
1006	4	5	1	2025-02-26
1007	5	5	1	2025-03-01
1008	5	1	1	2025-03-03
NULL	NULL	NULL	NULL	NULL

purchases 3

2.Use INNER, LEFT, and RIGHT JOIN to:

● Combine learner, course, and purchase data

● Display: Learner name, Course name, Category, Quantity, Total amount, Purchase date

● Presentation Requirements:

■ Format currency to 2 decimal places

■ Use column aliases

■ Sort by the highest total amount

- Inner join

learnears_id	courses_id	purchases_id	quantity	purchases_date	total_price
1	2	1002	1	2025-02-11	9,999.00
3	4	1005	1	2025-02-25	9,999.00
2	3	1003	1	2025-02-17	6,999.00
1	1	1001	1	2025-02-12	5,999.00
5	1	1008	1	2025-03-03	5,999.00
2	5	1004	1	2025-02-19	3,999.00
4	5	1006	1	2025-02-26	3,999.00
5	5	1007	0	2025-03-01	0.00

- Left join

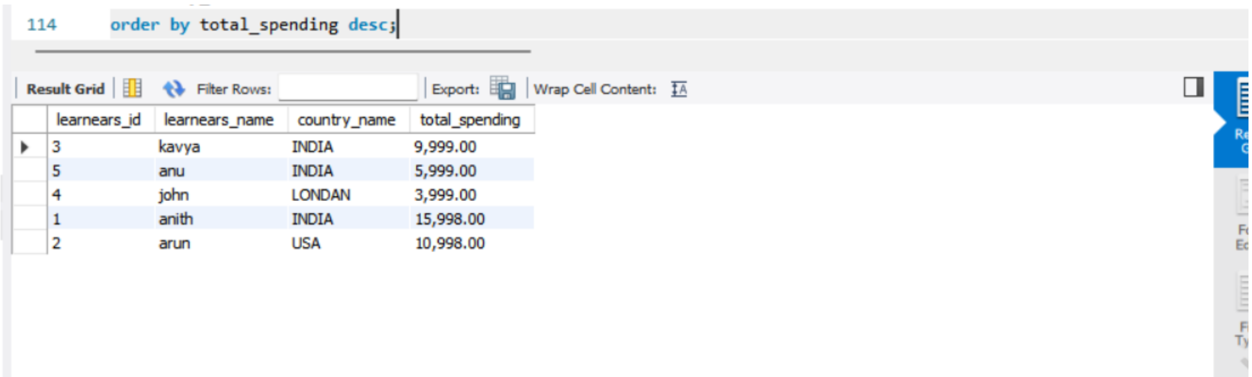
learnears_id	courses_id	purchases_id	quantity	purchases_date	total_price
1	2	1002	1	2025-02-11	9,999.00
3	4	1005	1	2025-02-25	9,999.00
2	3	1003	1	2025-02-17	6,999.00
1	1	1001	1	2025-02-12	5,999.00
5	1	1008	1	2025-03-03	5,999.00
2	5	1004	1	2025-02-19	3,999.00
4	5	1006	1	2025-02-26	3,999.00
5	5	1007	0	2025-03-01	0.00

- Right join

learnears_id	courses_id	purchases_id	quantity	purchases_date	total_price
1	2	1002	1	2025-02-11	9,999.00
3	4	1005	1	2025-02-25	9,999.00
2	3	1003	1	2025-02-17	6,999.00
1	1	1001	1	2025-02-12	5,999.00
5	1	1008	1	2025-03-03	5,999.00
2	5	1004	1	2025-02-19	3,999.00
4	5	1006	1	2025-02-26	3,999.00
5	5	1007	0	2025-03-01	0.00
NULL	6	NULL	NULL	NULL	NULL

3. **. Each learner's total spending with their country**

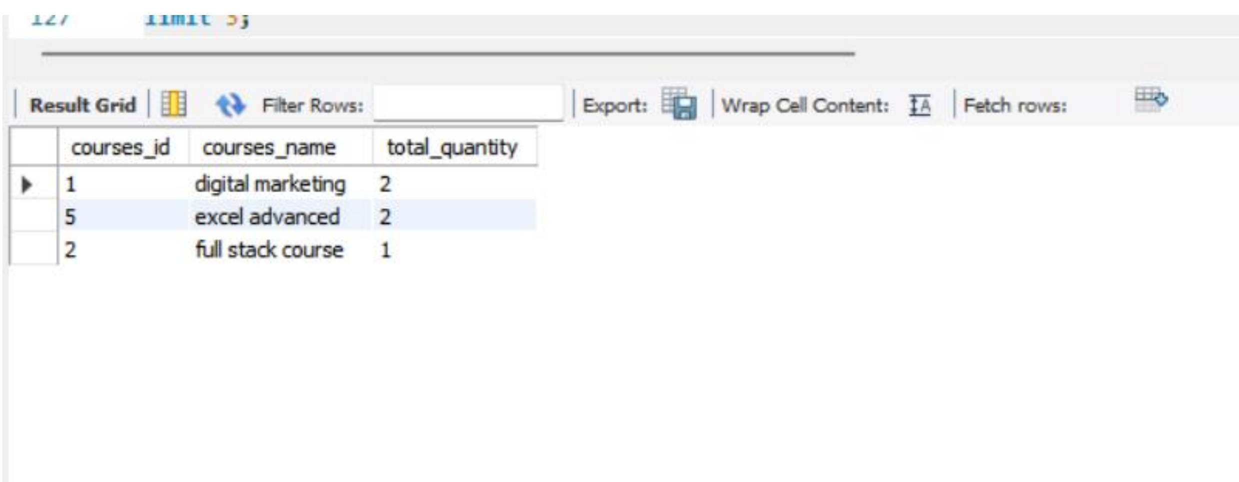
Q1. Display each learner's total spending with their country.



```
114    order by total_spending desc;
```

learnears_id	learnears_name	country_name	total_spending
3	kavya	INDIA	9,999.00
5	anu	INDIA	5,999.00
4	john	LONDAN	3,999.00
1	anith	INDIA	15,998.00
2	arun	USA	10,998.00

Q2. Find the top 3 most purchased courses by quantity.



```
127    limit 3;
```

courses_id	courses_name	total_quantity
1	digital marketing	2
5	excel advanced	2
2	full stack course	1

Q3. Show each category's:

- Total revenue
- Number of unique learners

category	total_revenue	unique_learnears
programming	19,998.00	2
marketing	11,998.00	2
productivity	7,998.00	3
data analytics	6,999.00	1

Q4. List learners who purchased from more than one category.

learnears_id	learnears_name
1	anith
2	arun
5	anu

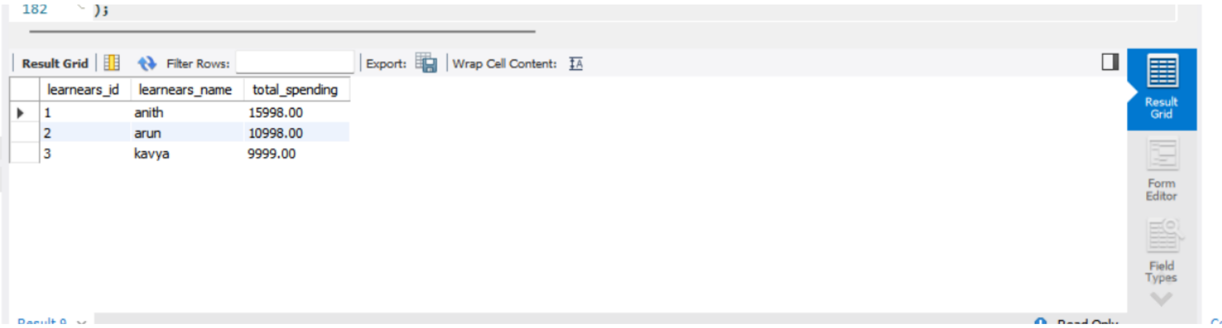
Q5. Identify courses never purchased.

```
(1003,02,003,1,'2025-02-17'),
```

courses_id	courses_name	category

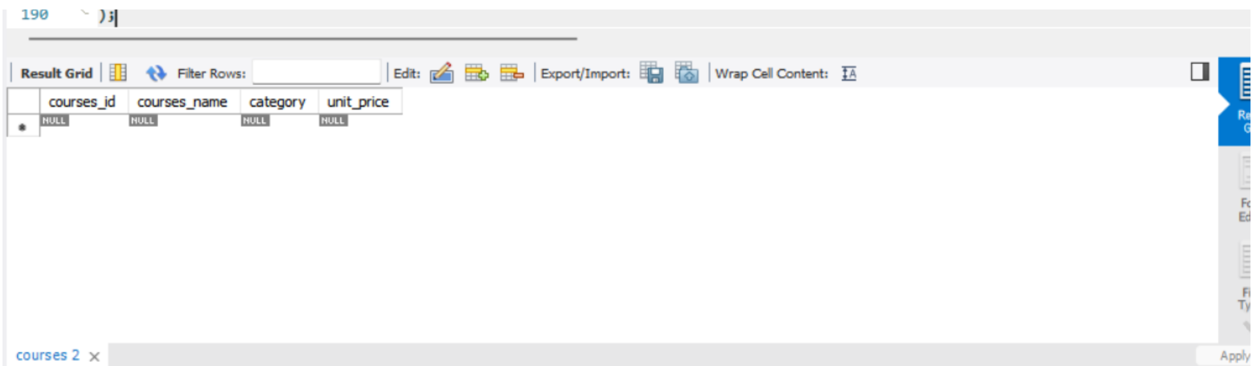
Result 22

Q6. Find learners whose total spending is above the average learner spending.



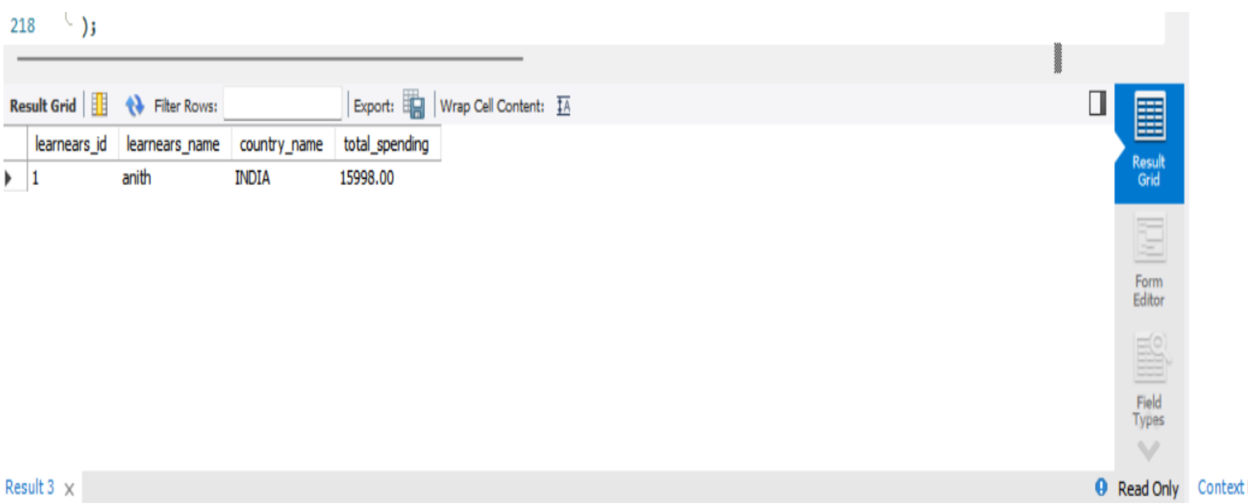
learnears_id	learnears_name	total_spending
1	anith	15998.00
2	arun	10998.00
3	kavya	9999.00

Q7. Display courses whose price is higher than any course in the ‘Beginner’ category.



courses_id	courses_name	category	unit_price
NULL	NULL	NULL	NULL

Q8 . Find learners who spent more than the average spending in their country.



learnears_id	learnears_name	country_name	total_spending
1	anith	INDIA	15998.00

5. CTE, CASE, View, and NULL Handling

Q9. Use a CTE to calculate total spending per learner, then:

Display learners with spending above 10,000.

learnears_id	learnears_name	total_spending
1	anith	15998.00
2	arun	10998.00

Q10. CASE Expression

Classify learners based on spending:

- Above 15,000 → "High Value",
- 8,000–15,000 → "Medium Value",
- Below 8,000 → "Low Value".

learnears_id	learnears_name	total_spending	learner_category
1	anith	15998.00	High Value
2	arun	10998.00	Medium Value
3	kavya	9999.00	Medium Value
4	john	3999.00	Low Value
5	anu	5999.00	Low Value

Q11 . NULL Handling

● Display all courses and replace NULL purchase counts with 0 using: IFNULL() or COALESCE()

courses_id	courses_name	purchase_count
1	digital marketing	2
2	full stack course	1
3	power bi quary	1
4	python	1
5	excel advanced	3
6	sql advanced	0

Q12 . View

● Create a view: category_performance_view

● Showing:

● Category

● Total revenue

● Number of purchases

● Average revenue per purchase

category	total_revenue	number_of_purchases	average_revenue_per_purchase
marketing	11998.00	2	5999.000000
programming	19998.00	2	9999.000000
data analytics	6999.00	1	6999.000000
productivity	7998.00	3	2666.000000

Conclusion

This SQL project helped us understand how to manage and analyze data using **JOINs, Subqueries, CTEs, CASE expressions, NULL handling, and Views**. We calculated learner spending, identified high-value learners, analyzed course and category performance, and created a reusable view for reporting. Overall, these SQL techniques help convert raw data into meaningful insights for better decision-making.